

TaskFlow AI – Customer Discovery Notes

Project: Taskflow – Email to Task Automation

Affiliation: Columbia University, COMS 6113 Agentic Systems Made Real

Prepared by Team 4: Blanca Valera Caballero, Diya Roongta, Emanuela Romano, Gautam Agarwal, Navinashok Swaminathan, Ron Zimmermann

Overview

This document summarizes the customer discovery and user research conducted by the TaskFlow AI team over the course of the project. The purpose of this work was to understand how users currently manage email-based action items, identify recurring pain points, and evaluate whether automated extraction of tasks and meetings from email could improve productivity. Insights were gathered through conversations with users, surveys, and hands-on evaluation of a functional prototype.

Problem & Rationale

Customer interviews and survey responses consistently highlighted email as the primary channel through which users receive tasks, deadlines, and meeting requests. Despite this, users reported relying on manual processes to identify and track action items, including rereading emails, copying content into task managers or calendars, or relying on memory. Several users described keeping emails unread as a proxy for 'to-do,' while others manually copied deadlines into separate tools. These approaches were described as inefficient and error-prone, often leading to missed follow-ups or delayed responses. This research validated the need for a solution that reduces manual triage while preserving user awareness and control.

Scope of Customer Research

The team engaged with undergraduate students, graduate students, working professionals, and researchers to capture a range of email usage patterns and productivity needs. Research activities included informal interviews, structured surveys, and direct observation of how users manage email-driven tasks. This scope allowed the team to compare behaviors across user segments and identify common challenges despite differences in roles and responsibilities.

Solution Concept Tested

Based on early customer insights, TaskFlow AI was developed as a functional prototype that extracts tasks and meeting details from emails and surfaces them for user review within existing Google workflows. We tested whether automating the conversion of emails into tasks and calendar events would address identified pain points. The concept focused on integrating directly with existing Gmail, Google Tasks, and Google Calendar workflows to minimize friction. User feedback during discovery emphasized the importance of transparency, leading to design choices that allow users to review generated tasks and events rather than relying on fully autonomous automation.

User Research Findings

Across interviews and survey responses, users reported that significant time is spent daily identifying actionable emails and transferring information into execution tools. Many participants noted frequent context switching and cognitive load associated with keeping track of tasks embedded in email threads. Users responded positively to the idea of automation when it reduced this overhead, particularly when they could see and edit the outcomes before finalizing them.

Validation & Evaluation

A user evaluation survey was conducted with 98 participants spanning students, professionals, and researchers (Figure 1). Results showed that approximately 60–70% of respondents reported spending less than 10 minutes per day on email triage and task identification when using TaskFlow. Satisfaction levels were high, with roughly 80% reporting satisfaction and an additional 20% reporting very high satisfaction. These findings supported the hypothesis that even partial automation could deliver meaningful efficiency gains.

What best describes you?

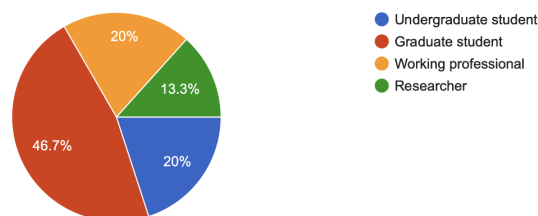


Figure 1

How often do you miss or forget a task or event because it gets buried in email?

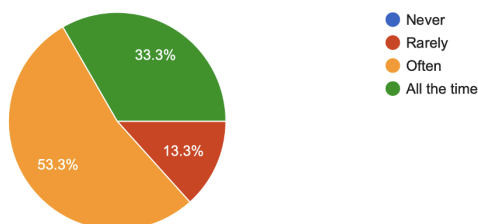


Figure 2

Survey results reinforced this pain point, with over 85% of respondents reporting that they often or consistently miss tasks because they get buried in email (Figure 2 & 3).

How often do you miss or forget a task or event because it gets buried in email now after TaskFlow?

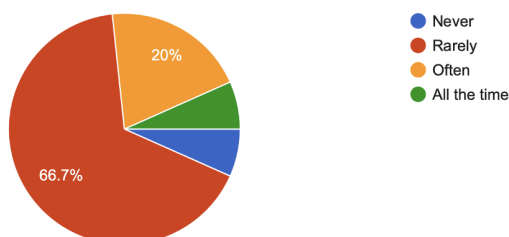


Figure 3

Improvement Suggestions from Users

In addition to validating the core problem and solution, customer discovery surfaced several recurring improvement suggestions. Many users expressed interest in a mobile implementation, noting that a significant portion of email triage and task review happens on smartphones rather than desktop browsers. Others highlighted the limitation of supporting only Google email accounts, suggesting that compatibility with non-Google providers would increase the product's relevance in professional and mixed-tool environments. Finally, some participants indicated a desire for deeper task execution support, such as progress tracking, reminders, or lightweight follow-up guidance, rather than stopping at task creation alone. These suggestions reflect unmet needs identified during discovery and point to opportunities for future exploration.

Key Insights

Customer discovery efforts revealed that the primary challenge is not email volume, but the lack of a seamless path from email to action. Users value solutions that integrate with tools they already use, require minimal behavior change, and provide reliability and control. Trust emerged as a central theme, reinforcing the importance of review mechanisms and predictable system behavior.

Conclusion

Customer research confirmed that manually extracting tasks and meetings from email is a widespread and recurring problem. TaskFlow AI was shaped directly by insights from user conversations and evaluations, and findings indicate strong alignment between user needs and the proposed solution. The discovery process validated both the problem and the direction of the product, providing a solid foundation for continued iteration and deeper customer engagement.